

AUGUSTINE PAUL SAMRAJ V

Backend Software Engineer | Node.js | TypeScript | Python | Microservices | Real-Time Systems

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PROFESSIONAL SUMMARY

Backend Software Engineer with nearly 4 years of experience building Node.js, TypeScript, and Python services for online assessment, live-proctoring, and fintech platforms. Skilled in REST APIs, microservices, WebSocket-based real-time pipelines, and Redis-backed caching and Pub/Sub systems that improved API response times by 30%+ on high-traffic endpoints. Built event-driven signaling (Firebase Realtime Database + WebSockets) in place of continuous video streaming to reduce dependence on video infrastructure while preserving live monitoring. Experienced in asynchronous job processing, MongoDB and PostgreSQL schema design, and API security (JWT, RBAC, rate limiting), with ownership of backend services end-to-end in Agile teams.

TECHNICAL SKILLS

Languages & Backend: Node.js, TypeScript, JavaScript, Python, Express.js, Microservices

APIs & Real-Time: REST APIs, WebSockets, Redis Pub/Sub, Firebase Realtime Database, Swagger/OpenAPI

Databases & Processing: MongoDB, PostgreSQL, Redis Caching, Bull Queue (Async Job Processing)

Security: JWT Authentication, RBAC Authorization, Request Validation, Rate Limiting, IP/Server Whitelisting

Testing & Observability: Jest, Mocha, k6 Load Testing, Pino, Postman

Cloud & Tooling: Docker, AWS, GCP, Git

PROFESSIONAL EXPERIENCE

Software Engineer (Backend) — DigiVal IT Solutions, Chennai, India

Sep 2022 - Present

Live Proctoring Platform (Digi Exam)

Feb 2026 - Present

- Architected a real-time event pipeline that converts third-party AI face-detection and head-pose outputs into structured anomaly events for backend processing and reporting.
- Designed a Firebase Realtime Database and WebSocket signaling layer as a low-latency alternative to continuous video streaming, reducing dependence on video infrastructure while preserving live monitoring.
- Built REST APIs for live-session management, recording, and proctor-event logging to support concurrent exam sessions at scale.
- Used Redis to cache session and presence state, reducing repeated database lookups during high-concurrency live-monitoring windows.
- Standardized face-not-detected, multiple-face, and gaze-deviation event schemas for consistent downstream violation tracking and reporting.

Infinity — Stock Market Analysis Platform (FinTech)

Jan 2025 - Feb 2026

- Designed Python-based backend microservices to ingest and process real-time stock-market data from third-party financial APIs under concurrent, low-latency conditions.
- Built a configurable price-target alert system that triggers notifications when stock prices cross user-defined thresholds.
- Implemented WebSocket-based live-price broadcasting with Redis Pub/Sub to distribute updates across concurrent client connections at scale.
- Modeled PostgreSQL schemas and optimized ingestion pipelines to maintain consistent low-latency delivery under concurrent user load.

Digi Assess & Digi Exam — Assessment Platforms

Sep 2022 - Jan 2025

- Built and maintained Node.js, TypeScript, and MongoDB REST APIs supporting end-to-end online examination workflows for more than two years.
- Improved API response times by approximately 30% on high-traffic assessment endpoints by implementing Redis caching and reducing repeated database access.
- Secured exam APIs with JWT authentication, RBAC authorization, request validation, rate limiting, and IP/server whitelisting for institution-restricted access.
- Implemented Bull Queue-based asynchronous report generation and email delivery, offloading long-running work from request handlers to improve API responsiveness and reliability.

EDUCATION

Bachelor of Computer Applications (BCA)

VELS Institute of Science, Technology and Advanced Studies (VISTAS)

CGPA: 7.9 / 10